🔑 **Answer Key — Teacher Copy · Lesson 3-1**

MSTAR PRACTICE WORKSHEET

Lesson 3-1: Understand Ratios

Grade 6 Mathematics · Standard 6.AT.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

A recipe uses 4 cups of milk and 1 cup of cream. What is the ratio of milk to cream?

A. 1:4

B. 4:1

C. 4:5

D. 5:1

A ratio compares quantities in the order given. Milk to cream = 4 to 1, or 4:1.

Question 2 SELECTED RESPONSE — CORRECT: A

Chef Reyes uses 6 strawberries and 2 bananas in a smoothie. What is the part-to-whole ratio of bananas to total fruit?

A. 2:8

B. 2:6

C. 6:2

D. 8:2

Total fruit = 6 + 2 = 8. The part-to-whole ratio of bananas to total fruit is 2:8.

Question 3 SELECTED RESPONSE — CORRECT: A

In a class of 30 students, 18 are girls. What is the part-to-part ratio of boys to girls?

A 12:18

B 12:30

C 18:12

D 18:30

Boys = 30 − 18 = 12. Part-to-part ratio of boys to girls = 12:18. (Order matters!)

Question 4 SELECTED RESPONSE — CORRECT: A

A pet store has 5 dogs and 9 cats. What is the ratio of dogs to cats?

A 5:9

B 5:14

C 9:5

D 14:5

The ratio of dogs to cats compares dogs first, then cats: 5 to 9, or 5:9.

Question 5 SELECTED RESPONSE — CORRECT: A

A fruit salad has 5 pieces of melon, 3 pieces of pineapple, and 2 grapes. What is the ratio of pineapple to melon?

A 3:5

B 3:10

C 3:2

D 5:3

The ratio of pineapple to melon compares pineapple (3) to melon (5), so the ratio is 3:5.

Question 6 SELECTED RESPONSE — CORRECT: A

A parking lot has 10 cars and 6 trucks. What is the part-to-whole ratio of trucks to total vehicles?

- (A) 6:16
- (B) 6:10
- (C) 10:6
- (D) 16:6

Total vehicles = 10 + 6 = 16. The ratio of trucks to total is 6:16.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: A

A class has 12 sixth graders and 18 seventh graders. What is the ratio of sixth graders to the TOTAL number of students, written in simplest form?

- (A) 2 : 5
- (B) 2 : 3
- (C) 3 : 5
- (D) 12 : 18

The total is 12 + 18 = 30. The ratio 12 : 30 divides by 6 to give 2 : 5.

- **B:** That is sixth graders to SEVENTH graders. The question asks for the ratio to the total of 30 students.
- **C:** The total is right but the first term is not. 12 to 30 divides by 6, giving 2 to 5.
- **D:** That is part to part, and unsimplified. The total is 12 + 18 = 30, so the comparison is 12 to 30.

Part B — correct answer: A

Which statement explains why 2 : 3 is NOT the answer to Part A?

- (A) 2 : 3 is the part-to-part ratio of sixth graders to seventh graders, not part-to-whole.
- (B) 2 : 3 has not been simplified completely.
- (C) 2 : 3 reverses the order of the quantities.
- (D) 2 : 3 counts each student twice.

12 : 18 simplifies to 2 : 3, which compares the two subgroups to each other. Part A asked for a part compared to the whole group of 30.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about ratios:

- **A** The order of the quantities in a ratio matters.
- **B** A ratio can compare a part to a whole or a part to another part.
- **C** 3 : 4 and 6 : 8 describe the same relationship.
- **D** A ratio must always compare quantities with the same units.
- **E** A ratio can be written as 3 : 4, 3 to 4, or 3/4.

A, B, C, and E are true. D is false — a rate is a ratio comparing quantities with DIFFERENT units, such as miles per hour.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A bag of marbles has 8 red, 6 blue, and 4 green marbles. Write three different ratios using these quantities: one part-to-part ratio, one part-to-whole ratio, and explain the difference.

Model answer: Part-to-part: red to blue = 8:6. Part-to-whole: green to total = 4:18. The difference is that part-to-part compares two individual groups, while part-to-whole compares one group to the entire collection (8 + 6 + 4 = 18).

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.